Chapter Contents

Chapter 21 - The Laplace Transform

The previous chapters studied differential equations directly in the original independent variable. The Laplace transform introduces a different viewpoint: it converts a function of time into a function of a new variable s.

For a suitable function $f(t)$, write:

$$\mathcal{L}\{f(t)\} = F(s).$$

This chapter develops the **forward** transform. We will:

- define it as an improper integral
- track the values of s for which that integral converges
- derive a small table of standard transforms
- use linearity, shifting, differentiation, scaling, and division properties
- transform finite piecewise and periodic functions

Inverse transforms begin in Chapter 22.

From A Time Function To An S-Function

The Goal Of This Block

The goal is to understand every symbol in the Laplace-transform definition and why convergence must be checked rather than assumed.

The Definition

For a function $f(t)$ defined for $t \geq 0$, its Laplace transform is:

$$\boxed{F(s) = \mathcal{L}\{f(t)\} = \int_0^\infty e^{-st} f(t)\, dt}.$$

Here:

- t is the integration variable, usually representing time
- s is a parameter and is treated as constant during the integration
- $f(t)$ is the original function
- $F(s)$ is the transformed function
- e^{-st} is the transform kernel

The braces in $\mathcal{L}\{f(t)\}$ identify the function being transformed. They are not multiplication symbols.

The Integral Is Improper

The upper limit ∞ means that the definition is shorthand for a limit:

$$\boxed{\mathcal{L}\{f(t)\} = \lim_{R\to\infty} \int_0^R e^{-st} f(t)\, dt}.$$

The transform exists at a particular s only when this limit is finite.

A formula for $F(s)$ without its convergence condition is incomplete. The same function can converge for some s and diverge for others.

What The Kernel Does

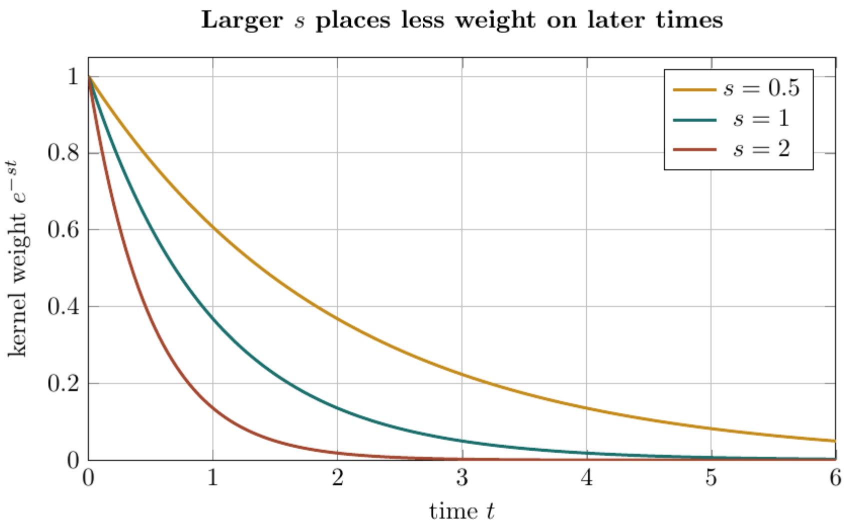

Figure 1: Exponential kernels for three positive values of s

For $s > 0$, the kernel e^{-st} decays as t increases. A larger s makes it decay faster, so late-time values of $f(t)$ receive less weight.

The key intuition:

> **$F(s)$ is a weighted summary of the entire future half-line $t \geq 0$, and changing s changes how strongly later times contribute.**

A First Transform From The Definition

Find the transform of:

$$f(t) = e^{-3t}.$$

Substitute $f(t)$ into the definition:

$$\begin{aligned} \mathcal{L}\{e^{-3t}\} &= \int_0^\infty e^{-st} e^{-3t}\, dt \\ &= \int_0^\infty e^{-(s+3)t}\, dt. \end{aligned}$$

Replace the improper integral by a limit:

$$\mathcal{L}\{e^{-3t}\} = \lim_{R\to\infty} \int_0^R e^{-(s+3)t}\, dt.$$

For $s + 3 \neq 0$, integrate with respect to t:

$$\begin{aligned} \int_0^R e^{-(s+3)t}\, dt &= \left[-\frac{e^{-(s+3)t}}{s+3} \right]_0^R \\ &= -\frac{e^{-(s+3)R}}{s+3} + \frac{1}{s+3}. \end{aligned}$$

The exponential term tends to zero exactly when:

$$s + 3 > 0,$$

or equivalently:

$$s > -3.$$

Since the remainder $e^{-(s+3)R}/(s+3)$ vanishes when $s > -3$, taking $R \to \infty$ gives:

$$\boxed{\mathcal{L}\{e^{-3t}\} = \frac{1}{s+3}, \qquad s > -3}.$$

Why The Other S-Values Fail

If $s = -3$, then the integrand becomes 1:

$$\int_0^R 1\,dt = R,$$

which diverges as $R \to \infty$.

If $s < -3$, then $-(s+3) > 0$, so $e^{-(s+3)t}$ grows rather than decays. The integral again diverges.

Thus the condition $s > -3$ follows from the limiting process; it is not an optional note attached after the calculation.

A Practical Existence Test

A common sufficient condition is:

- f is piecewise continuous on every finite interval $[0, R]$
- there are constants $M > 0$, a, and T such that

$$|f(t)| \le Me^{at} \qquad \text{for } t \ge T$$

Then f is said to be of **exponential order** a, and its Laplace transform converges for sufficiently large s, normally for $s > a$.

This condition is sufficient, not a claim that every transform has the same largest possible convergence region.

Block 1 Summary

The Laplace transform is an improper weighted integral:

$$F(s) = \int_0^\infty e^{-st} f(t)\, dt.$$

Treat s as a parameter, integrate with respect to t, evaluate the limit, and state the values of s for which it exists.

Building The Core Transform Table

The Goal Of This Block

The goal is to derive the transforms of constants, powers, and exponentials directly from the definition.

The Transform Of One

Start from the definition:

$$\mathcal{L}\{1\} = \lim_{R\to\infty} \int_0^R e^{-st}\, dt.$$

For $s \neq 0$:

$$\begin{aligned} \int_0^R e^{-st}\, dt &= \left[-\frac{1}{s} e^{-st}\right]_0^R \\ &= -\frac{e^{-sR}}{s} + \frac{1}{s}. \end{aligned}$$

When $s > 0$, $e^{-sR} \to 0$. Therefore:

$$\boxed{\mathcal{L}\{1\} = \frac{1}{s}, \qquad s > 0}.$$

At $s = 0$, the integral is $\int_0^\infty 1\, dt$, which diverges. For $s < 0$, the kernel grows.

The Transform Of T

Use integration by parts on the finite interval $[0, R]$:

$$u = t, \qquad dv = e^{-st}\,dt.$$

Differentiating $u = t$ and integrating $dv = e^{-st}\,dt$ gives:

$$du = dt, \qquad v = -\frac{1}{s}e^{-st}.$$

Apply $\int u\,dv = uv - \int v\,du$:

$$\begin{aligned}\int_0^R te^{-st}\,dt &= \left[-\frac{t}{s}e^{-st}\right]_0^R + \frac{1}{s}\int_0^R e^{-st}\,dt \\ &= -\frac{R}{s}e^{-sR} + \frac{1}{s}\left(-\frac{e^{-sR}}{s} + \frac{1}{s}\right).\end{aligned}$$

For $s > 0$, both e^{-sR} and Re^{-sR} tend to zero. Hence:

$$\boxed{\mathcal{L}\{t\} = \frac{1}{s^2}, \qquad s > 0}.$$

The Power Recurrence

Define:

$$I_n(s) = \int_0^\infty t^n e^{-st}\,dt.$$

For a positive integer n, integrate by parts with:

$$u = t^n, \qquad dv = e^{-st}\,dt.$$

Differentiating $u = t^n$ and integrating $dv = e^{-st}\,dt$ gives:

$$du = nt^{n-1}\,dt, \qquad v = -\frac{1}{s}e^{-st}.$$

The boundary term vanishes for $s > 0$, so:

$$\begin{aligned} I_n(s) &= \left[-\frac{t^n}{s}e^{-st}\right]_0^\infty + \frac{n}{s}\int_0^\infty t^{n-1}e^{-st}\,dt \\ &= \frac{n}{s}I_{n-1}(s). \end{aligned}$$

Since $I_0(s) = 1/s$, repeated use gives:

$$\begin{aligned} I_n(s) &= \frac{n}{s}\frac{n-1}{s}\cdots\frac{1}{s}I_0(s) \\ &= \frac{n!}{s^{n+1}}. \end{aligned}$$

Combining the recurrence with $I_0(s) = 1/s$ gives the power-transform rule:

$$\boxed{\mathcal{L}\{t^n\} = \frac{n!}{s^{n+1}}, \qquad s > 0}.$$

The symbol $n!$ means $n(n-1)\cdots 2 \cdot 1$.

A Power Example

For t^4, set $n = 4$ in the power formula:

$$\begin{aligned} \mathcal{L}\{t^4\} &= \frac{4!}{s^{4+1}} \\ &= \frac{4 \cdot 3 \cdot 2 \cdot 1}{s^5} \\ &= \frac{24}{s^5}, \qquad s > 0. \end{aligned}$$

Do not confuse the exponent 4 with the factorial 4! or the denominator power 5.

The General Exponential Transform

For $f(t) = e^{at}$:

$$\begin{aligned}\mathcal{L}\{e^{at}\} &= \int_0^\infty e^{-st}e^{at}\,dt \\ &= \int_0^\infty e^{-(s-a)t}\,dt.\end{aligned}$$

This is the transform of 1 with s replaced by $s - a$. Therefore:

$$\boxed{\mathcal{L}\{e^{at}\} = \frac{1}{s-a}, \qquad s > a}.$$

The sign in the denominator is opposite the sign in the exponent because:

$$e^{-st}e^{at} = e^{-(s-a)t}.$$

A Small Core Table

The results derived so far are:

$f(t)$	$\mathcal{L}\{f(t)\}$	Condition
1	$1/s$	$s > 0$
t^n	$n!/s^{n+1}$	$s > 0$
e^{at}	$1/(s-a)$	$s > a$

The condition column belongs to the transform pair. Keep it visible when the formula is derived from an improper integral.

Block 2 Summary

Integration by parts produces a recurrence for powers, while combining exponents gives the exponential transform. The basic table is small because later properties generate many new transforms from these entries.

Sine, Cosine, And Hyperbolic Functions

The Goal Of This Block

The goal is to derive the sine and cosine transforms as a coupled pair and then obtain hyperbolic transforms from exponentials.

Define Two Linked Integrals

For $s > 0$, define:

$$C(s) = \int_0^\infty e^{-st} \cos(bt)\, dt$$

and:

$$S(s) = \int_0^\infty e^{-st} \sin(bt)\, dt.$$

The exponential kernel ensures convergence because sine and cosine remain bounded.

First Integration By Parts

For $C(s)$, choose:

$$u = e^{-st}, \qquad dv = \cos(bt)\, dt.$$

Differentiating $u = e^{-st}$ and integrating $dv = \cos(bt)\,dt$ gives:

$$du = -se^{-st}\,dt, \qquad v = \frac{1}{b}\sin(bt).$$

The boundary term is zero for $s > 0$, so:

$$\begin{aligned} C(s) &= \left[\frac{1}{b}e^{-st}\sin(bt)\right]_0^\infty + \frac{s}{b}\int_0^\infty e^{-st}\sin(bt)\,dt \\ &= \frac{s}{b}S(s). \end{aligned}$$

Second Integration By Parts

For $S(s)$, choose:

$$u = e^{-st}, \qquad dv = \sin(bt)\,dt.$$

Differentiating $u = e^{-st}$ and integrating $dv = \sin(bt)\,dt$ gives:

$$du = -se^{-st}\,dt, \qquad v = -\frac{1}{b}\cos(bt).$$

Apply integration by parts:

$$\begin{aligned} S(s) &= \left[-\frac{1}{b}e^{-st}\cos(bt)\right]_0^\infty - \frac{s}{b}\int_0^\infty e^{-st}\cos(bt)\,dt \\ &= \frac{1}{b} - \frac{s}{b}C(s). \end{aligned}$$

The boundary term equals $1/b$ because its value at infinity is 0 and its value at $t = 0$ is $-1/b$.

Solve The Two Equations

Use:

$$C(s) = \frac{s}{b}S(s)$$

inside:

$$S(s) = \frac{1}{b} - \frac{s}{b}C(s).$$

Substitute and simplify one step at a time:

$$\begin{aligned} S(s) &= \frac{1}{b} - \frac{s}{b}\left(\frac{s}{b}S(s)\right) \\ &= \frac{1}{b} - \frac{s^2}{b^2}S(s). \end{aligned}$$

Add $(s^2/b^2)S(s)$ to both sides:

$$\left(1 + \frac{s^2}{b^2}\right)S(s) = \frac{1}{b}.$$

Write the factor with one denominator:

$$\frac{s^2 + b^2}{b^2}S(s) = \frac{1}{b}.$$

Multiply by $b^2/(s^2 + b^2)$:

$$S(s) = \frac{b}{s^2 + b^2}.$$

Substitute the solved value of $S(s)$ into $C(s) = (s/b)S(s)$:

$$\begin{aligned} C(s) &= \frac{s}{b}\frac{b}{s^2+b^2} \\ &= \frac{s}{s^2+b^2}. \end{aligned}$$

The solved functions $S(s)$ and $C(s)$ give the two transform formulas:

$$\boxed{\mathcal{L}\{\sin(bt)\} = \frac{b}{s^2+b^2}, \qquad s>0}$$

and:

$$\boxed{\mathcal{L}\{\cos(bt)\} = \frac{s}{s^2+b^2}, \qquad s>0}.$$

How To Remember The Numerators

The denominators match:

$$s^2+b^2.$$

The sine transform has b in the numerator, while the cosine transform has s:

$$\sin(bt) \longrightarrow \frac{b}{s^2+b^2}, \qquad \cos(bt) \longrightarrow \frac{s}{s^2+b^2}.$$

Check dimensions or set $b = 0$: $\cos(0t) = 1$ should give $1/s$, which confirms the cosine numerator.

Hyperbolic Functions From Exponentials

Use:

$$\cosh(bt) = \frac{e^{bt} + e^{-bt}}{2}.$$

Applying the exponential transform and combining fractions gives:

$$\begin{aligned}\mathcal{L}\{\cosh(bt)\} &= \frac{1}{2}\left(\frac{1}{s-b} + \frac{1}{s+b}\right) \\ &= \frac{1}{2}\left(\frac{(s+b)+(s-b)}{s^2-b^2}\right) \\ &= \frac{s}{s^2-b^2}.\end{aligned}$$

Similarly, because:

$$\sinh(bt) = \frac{e^{bt} - e^{-bt}}{2},$$

we obtain:

$$\mathcal{L}\{\sinh(bt)\} = \frac{b}{s^2-b^2}.$$

For real b, both formulas converge for $s > |b|$.

Block 3 Summary

Sine and cosine form a linked integration-by-parts system. Hyperbolic sine and cosine follow from their exponential definitions. The signs in $s^2 + b^2$ and $s^2 - b^2$ distinguish circular and hyperbolic functions.

Linearity And Direct Piecewise Transforms

The Goal Of This Block

The goal is to transform combinations term by term and handle a finite piecewise function directly from the integral definition.

Linearity

If:

$$\mathcal{L}\{f(t)\} = F(s), \qquad \mathcal{L}\{g(t)\} = G(s),$$

then for constants a and b:

$$\boxed{\mathcal{L}\{af(t) + bg(t)\} = aF(s) + bG(s)}.$$

This follows from linearity of integration:

$$\int_0^\infty e^{-st}[af(t) + bg(t)]\,dt = a\int_0^\infty e^{-st}f(t)\,dt + b\int_0^\infty e^{-st}g(t)\,dt.$$

A Complete Linearity Example

Find:

$$\mathcal{L}\{2t^3 - 5\sin(4t) + 7e^{-2t}\}.$$

Apply linearity before substituting table entries:

$$\mathcal{L}\{2t^3 - 5\sin(4t) + 7e^{-2t}\} = 2\mathcal{L}\{t^3\} - 5\mathcal{L}\{\sin(4t)\} + 7\mathcal{L}\{e^{-2t}\}.$$

Using the power, sine, and exponential rows of the transform table gives:

$$\mathcal{L}\{t^3\} = \frac{3!}{s^4} = \frac{6}{s^4},$$

$$\mathcal{L}\{\sin(4t)\} = \frac{4}{s^2+16},$$

and:

$$\mathcal{L}\{e^{-2t}\} = \frac{1}{s+2}.$$

Substitute these three results:

$$\boxed{\mathcal{L}\{2t^3 - 5\sin(4t) + 7e^{-2t}\} = \frac{12}{s^4} - \frac{20}{s^2+16} + \frac{7}{s+2}}.$$

All three terms converge together for $s > 0$, the most restrictive of their individual conditions.

A Finite Piecewise Function

Let:

$$q(t) = \begin{cases} t, & 0 \le t < 2, \\ 0, & t \ge 2. \end{cases}$$

Because $q(t) = 0$ after $t = 2$, the transform reduces to a finite integral:

$$\mathcal{L}\{q(t)\} = \int_0^2 te^{-st}\,dt.$$

For this integrand, integration by parts gives the antiderivative:

$$\int te^{-st}\,dt = -\frac{t}{s}e^{-st} - \frac{1}{s^2}e^{-st}.$$

Evaluate at both bounds:

$$\begin{aligned}\mathcal{L}\{q(t)\} &= \left[-\frac{t}{s}e^{-st} - \frac{1}{s^2}e^{-st}\right]_0^2 \\ &= -\frac{2e^{-2s}}{s} - \frac{e^{-2s}}{s^2} + \frac{1}{s^2} \\ &= \frac{1 - e^{-2s}(1+2s)}{s^2}.\end{aligned}$$

Thus, for $s \neq 0$:

$$\boxed{\mathcal{L}\{q(t)\} = \frac{1 - e^{-2s}(1+2s)}{s^2}}.$$

At $s = 0$, return to the finite integral rather than substituting into the fraction:

$$\mathcal{L}\{q(t)\}\big|_{s=0} = \int_0^2 t\,dt = 2.$$

The apparent singularity at $s = 0$ is removable.

Common Linearity Mistakes

Linearity permits sums and constant multiples. It does not generally permit:

$$\mathcal{L}\{f(t)g(t)\} = \mathcal{L}\{f(t)\}\mathcal{L}\{g(t)\}.$$

That equation is usually false.

Also keep the whole argument of a trigonometric or exponential function visible. For example, the 4 in $\sin(4t)$ becomes both the numerator 4 and the squared denominator term 16.

Block 4 Summary

Linearity turns sums into sums of transforms. A piecewise function can always be handled by splitting the defining integral over the intervals on which its formula changes.

Properties That Generate New Transforms

The Goal Of This Block

The goal is to use one known transform $F(s)$ to generate transforms involving exponential factors, powers of t, scaled inputs, and division by t.

Exponential Shifting

Suppose:

$$\mathcal{L}\{f(t)\} = F(s).$$

Multiply the original function by e^{at} and return to the definition:

$$\begin{aligned}\mathcal{L}\{e^{at}f(t)\} &= \int_0^\infty e^{-st}e^{at}f(t)\,dt \\ &= \int_0^\infty e^{-(s-a)t}f(t)\,dt.\end{aligned}$$

The last integral is F evaluated at $s-a$:

$$\boxed{\mathcal{L}\{e^{at}f(t)\} = F(s-a)}.$$

This is a shift in the transform variable, not a shift in time.

A Complete Shift Example

Find:

$$\mathcal{L}\{e^{2t}(t^2+3)\}.$$

First transform the unshifted function $f(t) = t^2 + 3$:

$$\begin{aligned} F(s) &= \mathcal{L}\{t^2\} + 3\mathcal{L}\{1\} \\ &= \frac{2!}{s^3} + \frac{3}{s} \\ &= \frac{2}{s^3} + \frac{3}{s}. \end{aligned}$$

The factor e^{2t} means $a = 2$, so replace every s in $F(s)$ by $s - 2$:

$$\boxed{\mathcal{L}\{e^{2t}(t^2+3)\} = \frac{2}{(s-2)^3} + \frac{3}{s-2}, \qquad s > 2}.$$

The convergence boundary also shifts from $s > 0$ to $s - 2 > 0$.

Multiplication By T

Differentiate the defining integral with respect to s under suitable regularity and convergence assumptions:

$$\begin{aligned} F'(s) &= \frac{d}{ds}\int_0^\infty e^{-st} f(t)\,dt \\ &= \int_0^\infty \frac{\partial}{\partial s}\left(e^{-st} f(t)\right)\,dt \\ &= \int_0^\infty (-t)e^{-st} f(t)\,dt. \end{aligned}$$

The differentiated integral states $F'(s) = -\mathcal{L}\{tf(t)\}$. Solving this identity for the transform gives:

$$\boxed{\mathcal{L}\{tf(t)\} = -F'(s)}.$$

Repeating the operation gives:

$$\boxed{\mathcal{L}\{t^n f(t)\} = (-1)^n F^{(n)}(s)}.$$

The exponent n counts derivatives with respect to s.

A Two-Derivative Example

Find $\mathcal{L}\{t^2 \sin(2t)\}$.

Begin with:

$$F(s) = \mathcal{L}\{\sin(2t)\} = \frac{2}{s^2 + 4}.$$

Because $n = 2$:

$$\mathcal{L}\{t^2 \sin(2t)\} = F''(s).$$

Differentiate once:

$$\begin{aligned} F'(s) &= 2(-1)(s^2 + 4)^{-2}(2s) \\ &= -\frac{4s}{(s^2 + 4)^2}. \end{aligned}$$

Differentiate the product $-4s(s^2+4)^{-2}$:

$$\begin{aligned} F''(s) &= -4(s^2+4)^{-2} + (-4s)(-2)(s^2+4)^{-3}(2s) \\ &= -\frac{4}{(s^2+4)^2} + \frac{16s^2}{(s^2+4)^3}. \end{aligned}$$

Rewrite the first fraction over the common denominator $(s^2+4)^3$:

$$\begin{aligned} F''(s) &= \frac{-4(s^2+4) + 16s^2}{(s^2+4)^3} \\ &= \frac{12s^2 - 16}{(s^2+4)^3} \\ &= \frac{4(3s^2-4)}{(s^2+4)^3}. \end{aligned}$$

Thus:

$$\boxed{\mathcal{L}\{t^2 \sin(2t)\} = \frac{4(3s^2-4)}{(s^2+4)^3}, \qquad s > 0}.$$

Scaling The Independent Variable

Let $c > 0$. Start from:

$$\mathcal{L}\{f(ct)\} = \int_0^\infty e^{-st} f(ct)\, dt.$$

Use the substitution:

$$u = ct, \qquad t = \frac{u}{c}, \qquad dt = \frac{du}{c}.$$

After the substitution $u = ct$, the transform integral becomes:

$$\begin{aligned} \mathcal{L}\{f(ct)\} &= \frac{1}{c}\int_0^\infty e^{-(s/c)u} f(u)\,du \\ &= \frac{1}{c}F\left(\frac{s}{c}\right). \end{aligned}$$

This change of variable proves the scaling rule:

$$\boxed{\mathcal{L}\{f(ct)\} = \frac{1}{c}F\left(\frac{s}{c}\right), \qquad c > 0}.$$

Both the outside factor $1/c$ and the replacement $s \mapsto s/c$ are needed.

Division By T

If $F(s) = \mathcal{L}\{f(t)\}$ and the required integrals converge, then:

$$\boxed{\mathcal{L}\left\{\frac{f(t)}{t}\right\} = \int_s^\infty F(u)\,du}.$$

The variable u is a dummy transform variable. It prevents confusion with the lower limit s.

As an example, use:

$$\mathcal{L}\{\sin(5t)\} = \frac{5}{s^2 + 25}.$$

Insert $F(u) = 5/(u^2 + 25)$ into the division-by-t formula:

$$\begin{aligned} \mathcal{L}\left\{\frac{\sin(5t)}{t}\right\} &= \int_s^\infty \frac{5}{u^2 + 25}\,du \\ &= \left[\arctan\left(\frac{u}{5}\right)\right]_s^\infty \\ &= \frac{\pi}{2} - \arctan\left(\frac{s}{5}\right). \end{aligned}$$

For $s > 0$, the complementary-angle identity gives:

$$\boxed{\mathcal{L}\left\{\frac{\sin(5t)}{t}\right\} = \arctan\left(\frac{5}{s}\right)}.$$

Although the expression contains $1/t$, the original function has a finite limit 5 as $t \to 0$.

Block 5 Summary

The four property patterns are:

$$\begin{aligned} e^{at}f(t) &\longrightarrow F(s-a), \\ t^n f(t) &\longrightarrow (-1)^n F^{(n)}(s), \\ f(ct) &\longrightarrow \frac{1}{c}F(s/c), \\ \frac{f(t)}{t} &\longrightarrow \int_s^\infty F(u)\,du. \end{aligned}$$

Each property changes a known transform in a specific way. Do not combine the rules by visual guesswork.

Periodic Functions

The Goal Of This Block

The goal is to reduce an integral over infinitely many repeated periods to one integral over a single period.

The Periodic Transform Formula

Suppose $p(t)$ has period $T > 0$:

$$p(t+T) = p(t).$$

Split the transform into period-length intervals:

$$\mathcal{L}\{p(t)\} = \sum_{k=0}^{\infty} \int_{kT}^{(k+1)T} e^{-st} p(t)\, dt.$$

In the kth interval, set $t = u + kT$. Periodicity gives $p(u+kT) = p(u)$, so:

$$\begin{aligned}\int_{kT}^{(k+1)T} e^{-st} p(t)\, dt &= \int_0^T e^{-s(u+kT)} p(u)\, du \\ &= e^{-skT} \int_0^T e^{-su} p(u)\, du.\end{aligned}$$

Factor out the one-period integral:

$$\mathcal{L}\{p(t)\} = \left(\int_0^T e^{-su} p(u)\, du \right) \sum_{k=0}^{\infty} e^{-skT}.$$

For $s > 0$, the geometric series is:

$$\sum_{k=0}^{\infty} e^{-skT} = \frac{1}{1 - e^{-sT}}.$$

Since the repeated periods contribute the geometric factor $\sum_{k=0}^{\infty} e^{-skT} = 1/(1-e^{-sT})$, multiplying by the one-period integral gives:

$$\boxed{\mathcal{L}\{p(t)\} = \frac{\int_0^T e^{-st} p(t)\,dt}{1-e^{-sT}}}.$$

An Original Periodic Ramp

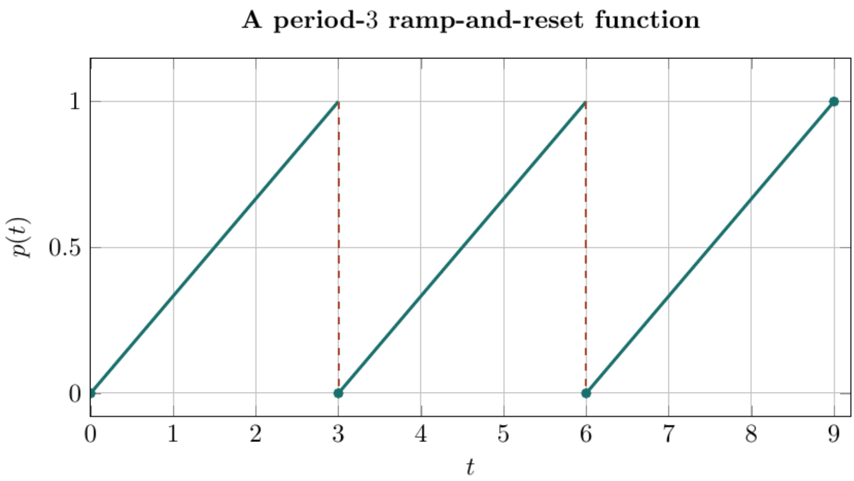

Figure 2: A period-three ramp-and-reset function

Define one period by:

$$p(t) = \frac{t}{3}, \qquad 0 \le t < 3,$$

and repeat it with period $T = 3$.

The jumps at $t = 3, 6, 9, \ldots$ do not prevent the transform from existing. The function is piecewise continuous and bounded.

Transform The Periodic Ramp

Use the periodic formula with $T = 3$:

$$\mathcal{L}\{p(t)\} = \frac{\int_0^3 e^{-st}\frac{t}{3}\,dt}{1-e^{-3s}}.$$

Factor out $1/3$:

$$\int_0^3 e^{-st}\frac{t}{3}\,dt = \frac{1}{3}\int_0^3 te^{-st}\,dt.$$

Use:

$$\int te^{-st}\,dt = -\frac{t}{s}e^{-st} - \frac{1}{s^2}e^{-st}.$$

Evaluate from 0 to 3:

$$\begin{aligned}\frac{1}{3}\int_0^3 te^{-st}\,dt &= \frac{1}{3}\left[-\frac{t}{s}e^{-st} - \frac{1}{s^2}e^{-st}\right]_0^3 \\ &= \frac{1}{3}\left(-\frac{3e^{-3s}}{s} - \frac{e^{-3s}}{s^2} + \frac{1}{s^2}\right) \\ &= \frac{1-e^{-3s}(1+3s)}{3s^2}.\end{aligned}$$

Substitute this one-period integral into the periodic formula:

$$\boxed{\mathcal{L}\{p(t)\} = \frac{1-e^{-3s}(1+3s)}{3s^2(1-e^{-3s})}, \qquad s > 0}.$$

Why One Period Is Enough

Every later period has the same shape, but the kernel contributes an extra factor:

$$1, \quad e^{-sT}, \quad e^{-2sT}, \quad e^{-3sT}, \quad \dots$$

The denominator $1 - e^{-sT}$ is the sum of those repeated exponential weights. It does not come from integrating the one-period formula itself.

Common Periodic-Function Mistakes

Do not:

- use the wrong period in e^{-sT}
- integrate over several periods and still divide by $1 - e^{-sT}$
- forget a piece of the one-period definition
- assume a jump discontinuity prevents a transform
- replace $p(t)$ by its formula from the first period outside that period without declaring periodic extension

Block 6 Summary

For a period-T function, transform one complete period and divide by the geometric-series factor $1 - e^{-sT}$. The numerator contains the shape; the denominator accounts for repetition.

A Reliable Transform Workflow

The Goal Of This Block

The goal is to choose the simplest valid route and audit the result before it is used in later differential-equation work.

Choose The Route Before Calculating

Ask these questions in order:

1. Is the function already a basic table entry?
2. Is it a sum or constant multiple suited to linearity?
3. Does it contain e^{at}, t^n, $f(ct)$, or $f(t)/t$?
4. Is it piecewise with finite support?
5. Is it periodic?
6. If none applies cleanly, should the definition be used directly?

This prevents a short property calculation from being replaced by several pages of unnecessary integration.

Core Table For This Chapter

$f(t)$	$F(s) = \mathcal{L}\{f(t)\}$
1	$1/s$
t^n	$n!/s^{n+1}$
e^{at}	$1/(s-a)$
$\sin(bt)$	$b/(s^2+b^2)$
$\cos(bt)$	$s/(s^2+b^2)$
$\sinh(bt)$	$b/(s^2-b^2)$
$\cosh(bt)$	$s/(s^2-b^2)$

The table is a starting point. Conditions of convergence still follow from the original function and any shifts that have been applied.

Fast Sanity Checks

Before accepting a transform, check:

- **Linearity:** were all outside constants preserved?
- **Power:** did t^n produce $n!$ and denominator power $n+1$?
- **Trigonometry:** does sine have b and cosine have s in the numerator?
- **Shift:** did e^{at} produce $s-a$, not $s+a$?
- **Scaling:** are both $1/c$ and s/c present?
- **Periodicity:** is the denominator $1-e^{-sT}$ using the true period?
- **Convergence:** is the stated s-range compatible with growth of $f(t)$?

Block 7 Summary

Identify structure before calculating, apply one property at a time, and keep the original time function visible while checking signs, parameters, and the convergence region.

Original Practice Set

Problems 1 To 4: Definition And Core Table

Problem 1: Transform A Constant From The Definition. Use the improper-integral definition to find $\mathcal{L}\{7\}$ and state its convergence condition.

Problem 2: Find A Convergence Boundary. Use the definition to transform e^{4t} and determine exactly which real values of s give convergence.

Problem 3: Transform A Power. Find $\mathcal{L}\{t^5\}$ and expand the factorial.

Problem 4: Use Linearity. Find:

$$\mathcal{L}\{3t^2 - 2\sin(5t)\}.$$

Problems 5 To 8: Transform Properties

Problem 5: Apply An Exponential Shift. Find $\mathcal{L}\{e^{-3t}(t+2)\}$ and state a convergence condition.

Problem 6: Multiply By T. Use differentiation in s to find $\mathcal{L}\{te^{-2t}\}$.

Problem 7: Differentiate Twice. Find $\mathcal{L}\{t^2 \sin(2t)\}$ without integrating in t.

Problem 8: Scale The Input. Given $f(t) = \sin t$ and $F(s) = 1/(s^2+1)$, use the scaling property to find $\mathcal{L}\{f(3t)\}$.

Problems 9 To 12: Division, Pieces, And Periodicity

Problem 9: Divide By T. Use the division property to find $\mathcal{L}\{\sin(5t)/t\}$ for $s > 0$.

Problem 10: Transform A Finite Pulse. Let:

$$r(t) = \begin{cases} 1, & 0 \le t < 2, \\ 0, & t \ge 2. \end{cases}$$

Find $\mathcal{L}\{r(t)\}$ directly from the definition.

Problem 11: Transform A Periodic Pulse. A period-3 function equals 2 on $0 \le t < 1$ and 0 on $1 \le t < 3$. Find its Laplace transform.

Problem 12: Diagnose A Shift Error. A learner writes:

$$\mathcal{L}\{e^{3t}f(t)\} = F(s+3).$$

Explain the sign error, give the correct formula, and describe how a boundary $s > \alpha$ for $F(s)$ changes.

Worked Answers: Problems 1 To 4

Answer 1. Start from the definition:

$$\begin{aligned}
\mathcal{L}\{7\} &= \lim_{R\to\infty} \int_0^R 7e^{-st}\,dt \\
&= 7 \lim_{R\to\infty} \left[-\frac{1}{s}e^{-st}\right]_0^R \\
&= 7 \lim_{R\to\infty} \left(-\frac{e^{-sR}}{s} + \frac{1}{s}\right).
\end{aligned}$$

For $s > 0$, the exponential tends to zero. Therefore:

$$\boxed{\mathcal{L}\{7\} = \frac{7}{s}, \qquad s > 0}.$$

Answer 2. Combine the exponentials:

$$\begin{aligned}
\mathcal{L}\{e^{4t}\} &= \int_0^\infty e^{-st}e^{4t}\,dt \\
&= \int_0^\infty e^{-(s-4)t}\,dt.
\end{aligned}$$

The kernel decays when $s - 4 > 0$. Hence:

$$\boxed{\mathcal{L}\{e^{4t}\} = \frac{1}{s-4}, \qquad s > 4}.$$

At $s = 4$ the integrand is 1; for $s < 4$ it grows.

Answer 3. Use the power formula with $n = 5$:

$$\begin{aligned}
\mathcal{L}\{t^5\} &= \frac{5!}{s^6} \\
&= \frac{5 \cdot 4 \cdot 3 \cdot 2 \cdot 1}{s^6} \\
&= \boxed{\frac{120}{s^6}}, \qquad s > 0.
\end{aligned}$$

Answer 4. Apply linearity and then substitute the two table entries:

$$\begin{aligned}
\mathcal{L}\{3t^2 - 2\sin(5t)\} &= 3\frac{2!}{s^3} - 2\frac{5}{s^2 + 25} \\
&= \boxed{\frac{6}{s^3} - \frac{10}{s^2 + 25}}, \qquad s > 0.
\end{aligned}$$

Worked Answers: Problems 5 To 8

Answer 5. First transform $f(t) = t + 2$, obtaining $F(s) = 1/s^2 + 2/s$.

The factor e^{-3t} has $a = -3$, so replace s by $s - (-3) = s + 3$:

$$\boxed{\mathcal{L}\{e^{-3t}(t+2)\} = \frac{1}{(s+3)^2} + \frac{2}{s+3}, \qquad s > -3}.$$

Answer 6. Start from $F(s) = \mathcal{L}\{e^{-2t}\} = 1/(s+2)$.

Use $\mathcal{L}\{tf(t)\} = -F'(s)$:

$$\begin{aligned}\mathcal{L}\{te^{-2t}\} &= -\frac{d}{ds}(s+2)^{-1} \\ &= -(-(s+2)^{-2}) \\ &= \boxed{\frac{1}{(s+2)^2}}, \qquad s > -2.\end{aligned}$$

Answer 7. For $f(t) = \sin(2t)$, the base transform is $F(s) = 2/(s^2+4)$. Since multiplication by t^2 corresponds to $F''(s)$:

$$\begin{aligned}F'(s) &= -\frac{4s}{(s^2+4)^2}, \\ F''(s) &= -\frac{4}{(s^2+4)^2} + \frac{16s^2}{(s^2+4)^3} \\ &= \frac{4(3s^2-4)}{(s^2+4)^3}.\end{aligned}$$

Because the required transform is $F''(s)$, the calculation gives:

$$\boxed{\mathcal{L}\{t^2\sin(2t)\} = \frac{4(3s^2-4)}{(s^2+4)^3}}.$$

Answer 8. Use the scaling property with $c = 3$:

$$\begin{aligned}
\mathscr{L}\{f(3t)\} &= \frac{1}{3}F\left(\frac{s}{3}\right) \\
&= \frac{1}{3}\frac{1}{(s/3)^2+1} \\
&= \frac{1}{3}\frac{9}{s^2+9} \\
&= \boxed{\frac{3}{s^2+9}}.
\end{aligned}$$

Since $f(3t) = \sin(3t)$, this agrees with the sine table entry.

Worked Answers: Problems 9 To 12

Answer 9. Use $F(u) = 5/(u^2 + 25)$ in the division property:

$$\begin{aligned}\mathcal{L}\left\{\frac{\sin(5t)}{t}\right\} &= \int_s^\infty \frac{5}{u^2+25}\,du \\ &= \left[\arctan\left(\frac{u}{5}\right)\right]_s^\infty \\ &= \frac{\pi}{2} - \arctan\left(\frac{s}{5}\right) \\ &= \boxed{\arctan\left(\frac{5}{s}\right)}, \qquad s > 0.\end{aligned}$$

Answer 10. The function vanishes after $t = 2$, so:

$$\begin{aligned}\mathcal{L}\{r(t)\} &= \int_0^2 e^{-st}\,dt \\ &= \left[-\frac{1}{s}e^{-st}\right]_0^2 \\ &= \boxed{\frac{1-e^{-2s}}{s}}\end{aligned}$$

for $s \neq 0$. At $s = 0$, the finite integral equals 2.

Answer 11. The period is $T = 3$. Transform one period:

$$\begin{aligned}\int_0^3 e^{-st}p(t)\,dt &= \int_0^1 2e^{-st}\,dt + \int_1^3 0\,dt \\ &= \frac{2(1-e^{-s})}{s}.\end{aligned}$$

For Answer 11, the one-period integral is $2(1-e^{-s})/s$. The periodic-transform formula divides this result by $1-e^{-3s}$:

$$\boxed{\mathcal{L}\{p(t)\} = \frac{2(1-e^{-s})}{s(1-e^{-3s})}, \qquad s > 0}.$$

Answer 12. Combine the kernels before identifying the shift:

$$e^{-st}e^{3t} = e^{-(s-3)t}.$$

Therefore the correct property is:

$$\boxed{\mathcal{L}\{e^{3t}f(t)\} = F(s-3)}.$$

If $F(s)$ originally converges for $s > \alpha$, the shifted argument must satisfy:

$$s - 3 > \alpha.$$

Add 3 to both sides:

$$\boxed{s > \alpha + 3}.$$

Chapter 21 Final Summary

The defining formula and the periodic-function rule are:

$$\boxed{\mathcal{L}\{f(t)\} = \int_0^\infty e^{-st} f(t)\,dt}, \qquad \boxed{\mathcal{L}\{p(t)\} = \frac{\int_0^T e^{-st} p(t)\,dt}{1 - e^{-sT}}}.$$

The core table entries and transform-building properties are:

$$\begin{aligned}
1 &\longrightarrow \frac{1}{s}, & t^n &\longrightarrow \frac{n!}{s^{n+1}}, & e^{at} &\longrightarrow \frac{1}{s-a}, \\
\sin(bt) &\longrightarrow \frac{b}{s^2+b^2}, & \cos(bt) &\longrightarrow \frac{s}{s^2+b^2}, \\
e^{at} f(t) &\longrightarrow F(s-a), & t^n f(t) &\longrightarrow (-1)^n F^{(n)}(s), & f(ct) &\longrightarrow \frac{1}{c} F(s/c).
\end{aligned}$$

Together, these results support one central working principle:

> **identify the time-domain structure, apply one justified property at a time, and carry the convergence condition with the transform rather than treating it as an afterthought.**